



Name: Cohort/Batch: Date: Score:

AZURE SYNAPSE PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Data Platforms

TOTAL: 10 QUESTIONS

Q1 What is Azure Synapse and what problem in Data Platforms does it solve?

Q6 How does Azure Synapse handle reliability and high availability?

Q2 What are the core components or concepts in Azure Synapse?

Q7 What observability does Azure Synapse provide out of the box?

Q3 How does Azure Synapse compare to its main alternatives?

Q8 What are common performance bottlenecks in Azure Synapse?

Q4 How do you get started with Azure Synapse for a new project?

Q9 What security best practices apply when running Azure Synapse?

Q5 What configuration options most affect Azure Synapse's behavior in production?

Q10 What mistakes do teams commonly make when adopting Azure Synapse?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Azure Synapse is a tool or platform

Mitigation

Azure Synapse is a tool or platform that automates or simplifies a specific concern in the Data Platforms domain, reducing manual effort and operational complexity for engineering teams.

A6 Azure Synapse provides HA through leader election, observability

Azure Synapse provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Azure Synapse has key building blocks that

Primitives

Azure Synapse has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Azure Synapse exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

Information

A3 Azure Synapse trades off simplicity against feature

Hardening

Azure Synapse trades off simplicity against feature richness differently than its alternatives. Choose Azure Synapse when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Azure Synapse via its package manager

Gotchas

Install Azure Synapse via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Azure Synapse with the principle of

Assessment

Run Azure Synapse with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency, generation

Concurrency

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and

Documentation

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.